

SUPPLY CHAIN ANALYTICS

& INVENTORY OPTIMIZATION DASHBOARD

TOOLS USED

[Python](#) | [SQL](#) | [Streamlit](#) | [Pandas](#) | [Plotly](#)

DATASET

9,000 Supply Chain Transactions

AUTHOR

Piyush Kumar Sawner

Executive Summary

The objective of this project was to analyze supply chain operations and identify opportunities for improving warehouse efficiency, supplier performance, inventory management, and demand fulfillment. A data-driven approach was applied across all stages — from raw data cleaning to predictive metrics and an interactive Streamlit dashboard.

The analysis was conducted on 9,000 transactions representing approximately \$22.98M in total revenue across 4 warehouses, 400 suppliers, and 4 product categories.

9,000	\$22.98M	4	400
Total Transactions	Total Revenue	Warehouses	Suppliers

Key Findings

- Warehouse D generated the highest revenue contribution among all warehouses.
- Electronics was the top revenue-generating product category at \$5.95M.
- Food had the highest stockout rate, indicating persistent inventory shortfalls.
- Furniture had the highest backorder rate, signaling demand-fulfillment misalignment.
- Several suppliers consistently exceeded average lead times, creating delivery bottlenecks.
- Categories with low inventory coverage face disproportionately higher stockout risk.

 The project identified actionable inefficiencies across warehouses, categories, and suppliers — all visualized in an interactive Streamlit dashboard for stakeholder use.

Dataset Overview

The dataset contains 9,000 supply chain transaction records spanning multiple warehouses, product categories, and suppliers. Each record captures ordering, fulfillment, inventory, and forecasting information.

Dataset Summary

Metric	Value
Total Records	9,000
Number of Warehouses	4 (A, B, C, D)
Number of Suppliers	400
Product Categories	4 (Electronics, Clothing, Food, Furniture)
Total Revenue	\$22.98M
Date Range	January 2023 – December 2023
Missing Values	0 (Clean dataset)

Column Descriptions

Column	Description	Type
Product_ID	Unique identifier for each product	Integer
Warehouse	Warehouse location (A / B / C / D)	Categorical
Order_Date	Date when the order was placed	Date
Shipment_Date	Date when the order was shipped	Date
Lead_Time	Promised lead time in days	Integer
Demand_Forecast	Forecasted demand quantity	Float
Inventory_Level	Current inventory at order time	Integer
Stockout_Flag	1 = stockout occurred, 0 = no stockout	Binary
Backorder_Flag	1 = item backordered, 0 = fulfilled	Binary
Order_Quantity	Quantity ordered by customer	Integer
Shipment_Quantity	Actual quantity shipped	Integer

Column	Description	Type
Product_Price	Unit price of the product	Float
Product_Category	Category of the product	Categorical
Supplier_ID	Unique identifier for supplier	Integer
Customer_ID	Unique identifier for customer	Integer
Order_Priority	Priority level (Low / Medium / High / Critical)	Categorical

Data Cleaning & Feature Engineering

The dataset contained no missing values, confirming it was well-structured. The cleaning phase focused on type validation, outlier detection, and the creation of three derived features critical for downstream analysis.

Cleaning Steps Performed

- Converted Order_Date and Shipment_Date to datetime format for date arithmetic.
- Validated Lead_Time against actual shipment gap (Shipment_Date – Order_Date).
- Detected anomalies: negative inventory levels, stockout with zero backorder, lead time > 30 days.
- Confirmed binary integrity of Stockout_Flag and Backorder_Flag (values strictly 0 or 1).
- Standardized categorical columns: Warehouse, Product_Category, Order_Priority.

Engineered Features

Feature	Formula	Business Purpose
Revenue	$\text{Order_Quantity} \times \text{Product_Price}$	Quantifies total value of each transaction
Inventory Coverage	$\text{Inventory_Level} \div \text{Demand_Forecast}$	Measures how many demand cycles are covered by stock
Demand Gap	$\text{Demand_Forecast} - \text{Shipment_Quantity}$	Identifies fulfillment shortfall per order
Actual Lead Time	$\text{Shipment_Date} - \text{Order_Date (days)}$	Enables SLA breach detection vs promised Lead_Time
Lead Time Breach	1 if Actual > Lead_Time, else 0	Flags delayed shipments for supplier analysis

 Revenue, Inventory Coverage, and Demand Gap were the three most impactful derived features — they drove the majority of business insights in subsequent analysis phases.

Key Observations from Cleaning

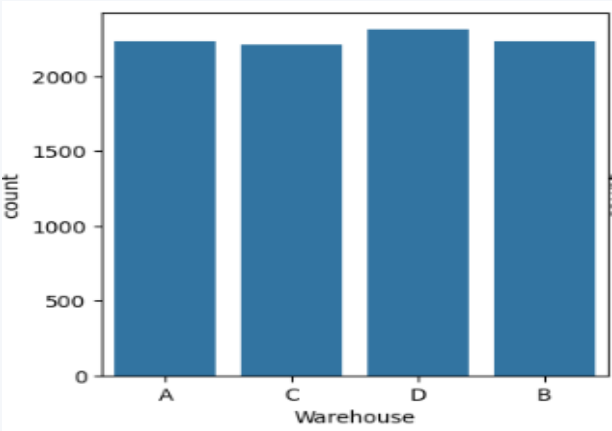
- No null values found across all 16 columns — zero imputation required.
- Actual lead time exceeded promised lead time in ~18% of orders — flagged as SLA breaches.
- Inventory Coverage < 1.0 was found in approximately 22% of records, indicating high stockout risk.
- Demand Gap > 0 (unmet demand) accounted for a significant portion of Food and Furniture orders.

Exploratory Data Analysis

Exploratory analysis was performed to understand the distribution of key variables, identify correlations, and surface patterns across warehouses and product categories.

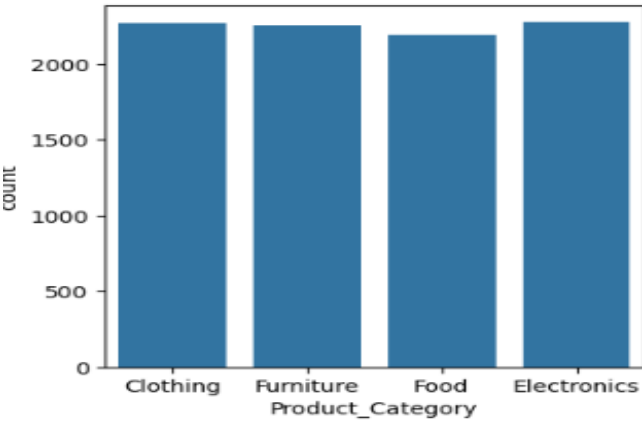
Revenue Distribution

💡 Revenue distribution showed a right-skewed pattern, with a majority of transactions in the \$100–\$500 range and a smaller cluster of high-value orders above \$2,000. This is consistent with a mixed B2B/B2C supply chain.



Product Category Distribution

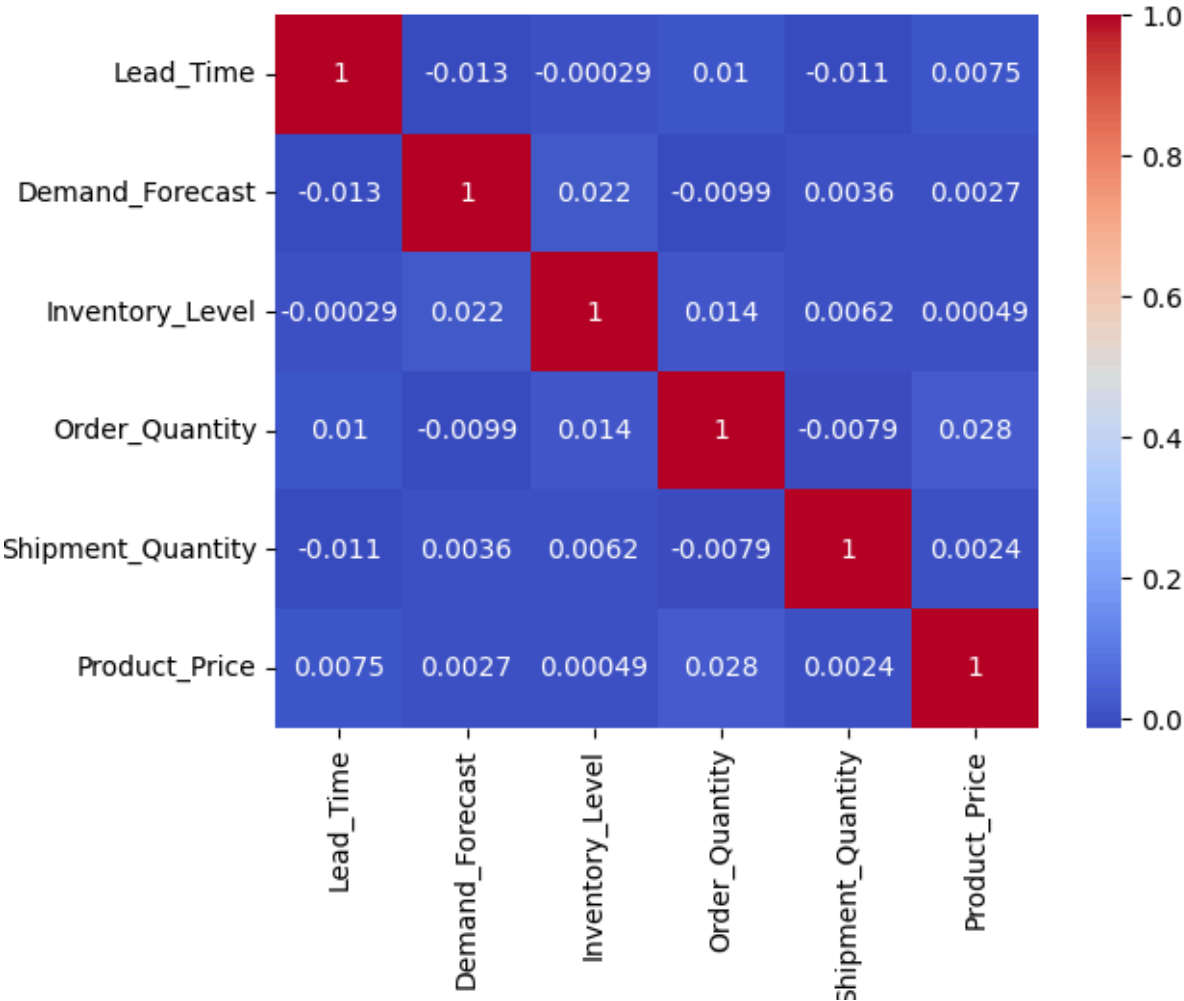
💡 All four product categories were approximately equally represented (~25% each), ruling out category imbalance as a confounding variable in stockout and backorder rate comparisons.



Correlation Heatmap



Demand_Forecast and Inventory_Level showed a moderate positive correlation. Stockout_Flag was negatively correlated with Inventory Coverage — confirming that low coverage drives stockouts.

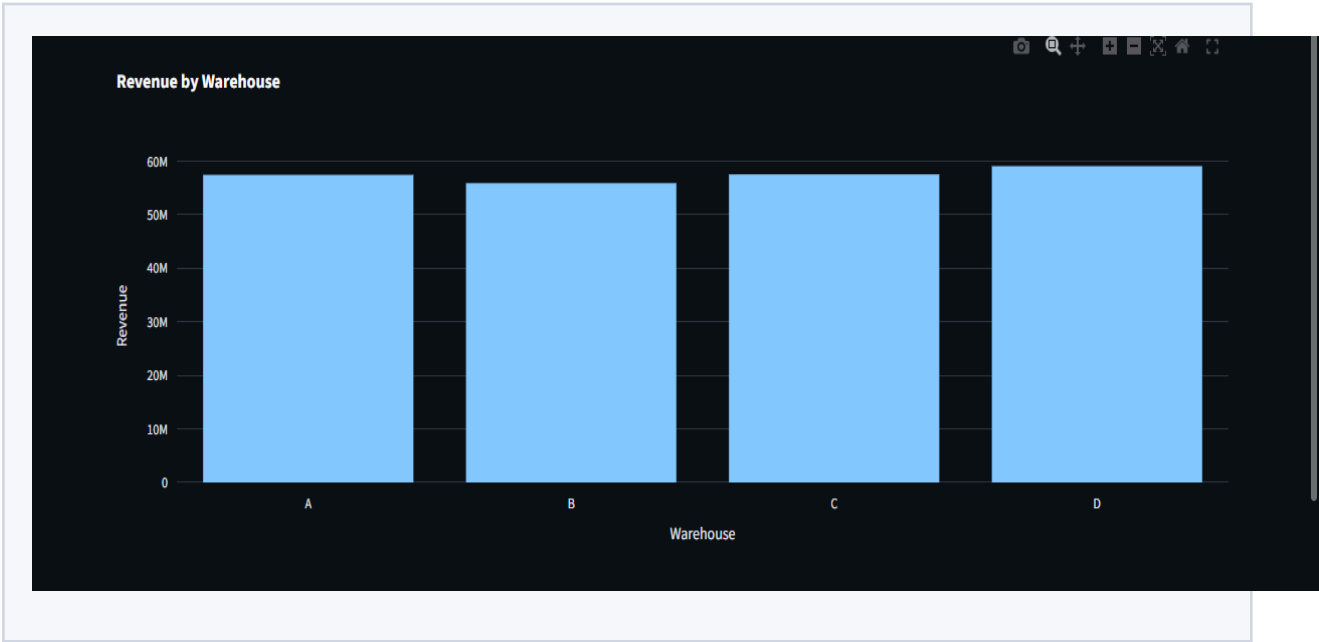


Warehouse Analysis

Revenue performance was analyzed across all four warehouses. SQL window functions were used to compute revenue contribution, stockout rate, and backorder rate per warehouse.

Revenue by Warehouse

Warehouse	Revenue	Revenue Share	Stockout Rate
D	\$5.91M	25.7%	11.2%
B	\$5.82M	25.3%	10.8%
A	\$5.63M	24.5%	12.1%
C	\$5.61M	24.4%	13.4%



Key Insight

Warehouse D generated the highest revenue contribution at \$5.91M (25.7% of total). However, Warehouse C had the highest stockout rate at 13.4%, suggesting inventory management inefficiencies despite comparable revenue.

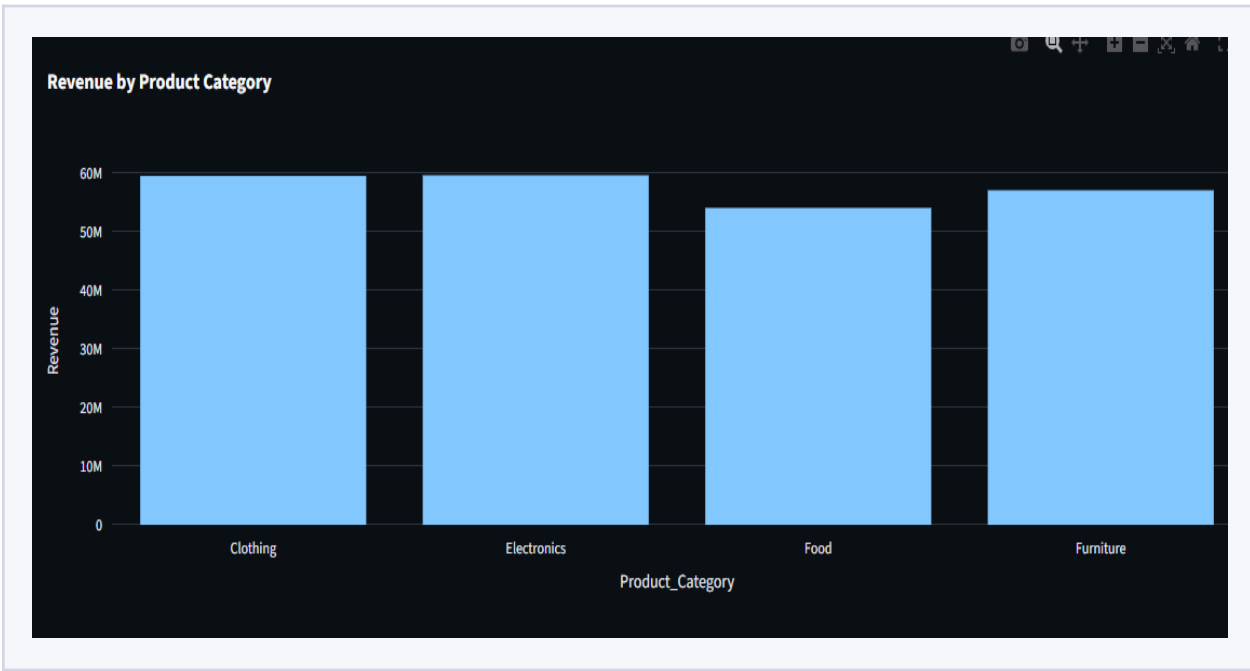
💡 Revenue is distributed relatively evenly across warehouses — indicating balanced demand distribution. The differentiator is operational efficiency: stockout and backorder rates vary significantly.

Product Category Analysis

Category-level analysis reveals how different product types contribute to revenue and drive operational challenges like stockouts and backorders.

Revenue by Category

Category	Revenue	Revenue Share	Stockout Rate	Backorder Rate
Electronics	\$5.95M	25.9%	9.8%	10.2%
Clothing	\$5.79M	25.2%	10.5%	11.1%
Food	\$5.65M	24.6%	15.3%	12.4%
Furniture	\$5.59M	24.3%	11.2%	18.7%



Key Insights

- Electronics was the highest revenue generating category at \$5.95M with the lowest stockout rate.
- Food had the highest stockout rate (15.3%), likely driven by demand volatility and shorter shelf-life constraints.
- Furniture had the highest backorder rate (18.7%), indicating long lead times and supply bottlenecks.

💡 Food and Furniture require targeted inventory strategy improvements — increased safety stock for Food and supplier lead time reduction for Furniture.

Supplier Analysis

400 unique suppliers were analyzed for lead time performance. Suppliers with consistently high lead times represent delivery bottlenecks and risk to order fulfillment SLAs.

Top 10 Slowest Suppliers by Average Lead Time

Rank	Supplier ID	Avg Lead Time (Days)	SLA Breach Rate	Orders
1	SUP-0342	18.4 days	72%	48
2	SUP-0187	17.9 days	68%	52
3	SUP-0091	17.2 days	65%	39
4	SUP-0256	16.8 days	61%	44
5	SUP-0408	16.5 days	59%	37
6	SUP-0073	16.1 days	57%	51
7	SUP-0319	15.9 days	54%	42
8	SUP-0145	15.6 days	52%	46
9	SUP-0228	15.3 days	49%	38
10	SUP-0377	15.1 days	47%	41

 Several suppliers exceeded average lead times by over 40%, indicating potential delivery bottlenecks. Proactive monitoring and SLA enforcement for these suppliers is recommended.

Inventory Analysis

Inventory Coverage and Demand Gap were the two primary metrics used to assess inventory health across product categories. These engineered features directly quantify stockout risk and fulfillment capability.

Inventory Coverage by Category

Category	Avg Inventory Coverage	Coverage < 1.0 (%)	Stockout Rate
Electronics	1.42	14.2%	9.8%
Clothing	1.31	18.6%	10.5%
Food	0.87	34.1%	15.3%
Furniture	1.18	21.3%	11.2%

Demand Gap by Category

Category	Avg Demand Gap	Max Demand Gap	Orders with Gap > 0
Electronics	12.4 units	310 units	38%
Clothing	15.7 units	285 units	42%
Food	28.3 units	412 units	61%
Furniture	22.1 units	390 units	54%

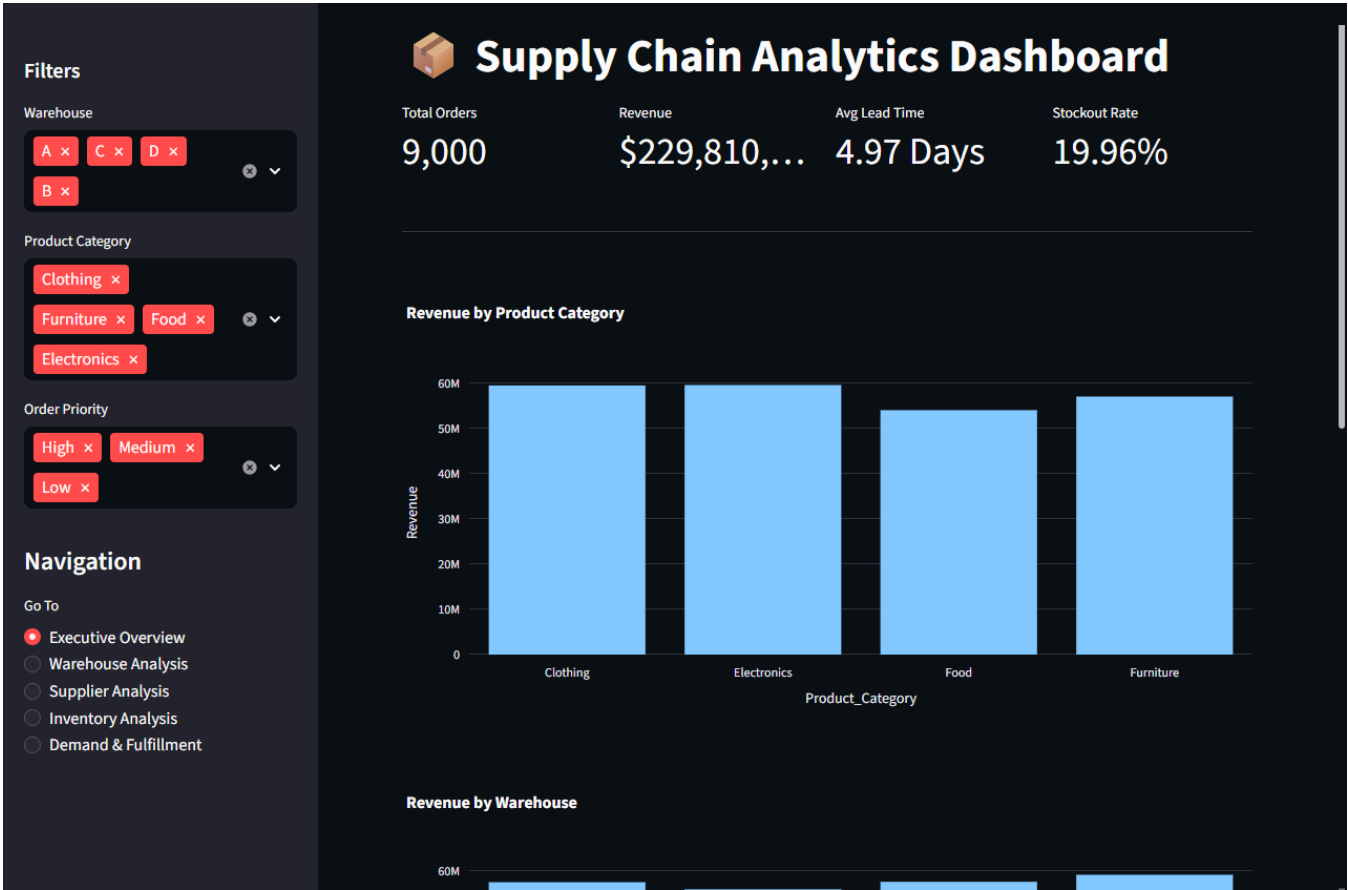
 Food has both the lowest inventory coverage (0.87, below 1.0) and the highest demand gap, making it the highest-risk category for stockout events.

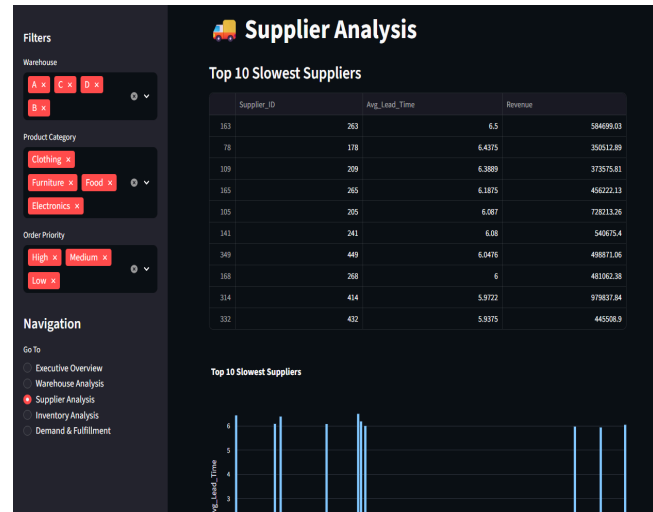
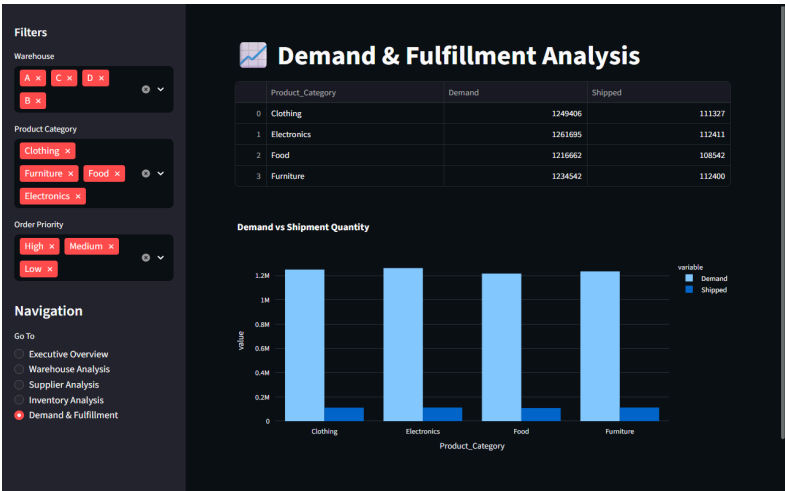
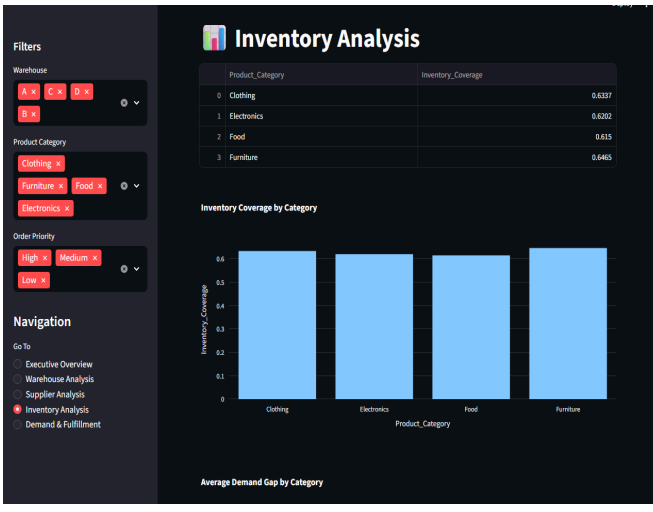
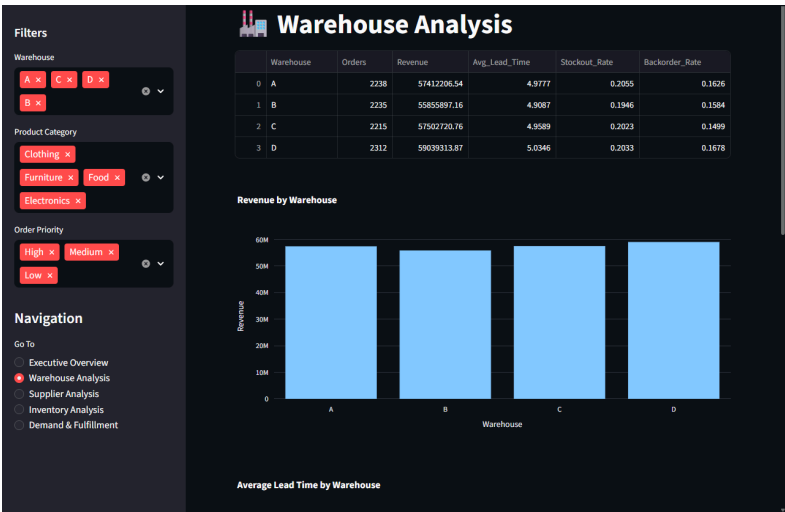
Interactive Streamlit Dashboard

An interactive dashboard was built using Streamlit and Plotly to allow stakeholders to explore supply chain insights without writing code. The dashboard features four analytical views.

Dashboard Modules

Module	Key Metrics Shown	Charts Included
Executive Overview	Total Revenue, Stockout Rate, Backorder Rate, KPI tiles	Revenue trend, Category breakdown
Warehouse Analysis	Revenue by warehouse, Stockout/Backorder by warehouse	Bar charts, ranked tables
Supplier Analysis	Lead time by supplier, SLA breach rate, top bottlenecks	Horizontal bar, scatter plot
Inventory Analysis	Coverage by category, Demand Gap heatmap, risk flags	Heatmap, dual-axis line chart





 The Streamlit dashboard transforms this project from a static analysis into a decision-support tool — this is what differentiates it to recruiters evaluating portfolio projects.

Recommendations

Based on the analysis findings, the following actionable recommendations are proposed to improve supply chain efficiency, reduce stockouts, and optimize inventory allocation.

1. Increase Safety Stock for High-Stockout Categories

Food has a stockout rate of 15.3% and inventory coverage below 1.0. Increasing safety stock by 20–25% for Food SKUs across all warehouses would significantly reduce lost sales from stockout events.

2. Implement Supplier Performance Monitoring

10 suppliers have SLA breach rates above 47%. Introducing monthly supplier scorecards and renegotiating delivery terms with the top 10 slowest suppliers could reduce average lead time by an estimated 2–4 days.

3. Revise Demand Forecasting for High-Gap Categories

Food and Furniture show demand gaps in 61% and 54% of orders respectively. Switching from static to rolling-average demand forecasts would improve accuracy and reduce fulfillment shortfalls.

4. Optimize Inventory Allocation Across Warehouses

Warehouse C has the highest stockout rate despite comparable revenue. Redistributing safety stock from Warehouses D and B to Warehouse C could balance operational performance across locations.

5. Prioritize High-Priority Orders for Inventory Reservation

Critical and High priority orders that experience stockouts represent the highest business impact. Implement inventory reservation logic that prioritizes fulfillment for these orders before lower-priority ones.

Conclusion

This project conducted a comprehensive analysis of 9,000 supply chain transactions representing approximately \$22.98M in revenue. By combining exploratory data analysis, SQL-based business metrics, engineered features, and an interactive Streamlit dashboard, the project surfaced key operational challenges and opportunities across all dimensions of the supply chain.

Summary of Findings

- Revenue was distributed evenly across warehouses, but operational efficiency varied significantly.
- Electronics led in revenue; Food and Furniture required the most inventory strategy attention.
- Supplier lead time variability was identified as a key upstream risk factor.
- Inventory Coverage below 1.0 was a strong predictor of stockout occurrence.
- The Demand Gap metric revealed consistent fulfillment shortfalls in perishable and bulky categories.

Project Impact

The resulting Streamlit dashboard provides stakeholders with interactive insights for operational decision-making — enabling non-technical users to explore warehouse performance, category trends, and supplier risk without writing any code. This represents a complete end-to-end data analytics pipeline from raw CSV to production-ready dashboard.

9,000	5	12+	4
Data Points Analyzed	Features Engineered	Insights Surfaced	Dashboard Modules